

UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES
STA5073Z – Data Science for Industry 2026

Assignment 2
Image classification with neural networks

This project must be done in groups of 2-3
Due date: 16 October

Problem background

You have been given a dataset of 25,000 handwritten digits written by 20 people. Each person completed five independent recording sessions, in each of which they were asked to provide 50 samples of the digits 0-4 i.e. $5 \times 50 \times 5 = 1250$ samples per person. Each person has a consistent handwriting style, but their writing changes somewhat between five recording sessions (perhaps they are given different writing materials, or it could be natural variation). Individual examples also vary within a session. The images are simulated for this assignment and are not real handwriting.

The data are in ‘handwriting.rds’ with details given below. You may use any programming language and neural network library for this assignment.

Tasks

Note that tasks have been designed to be performed on a standard laptop or desktop with a CPU. Models trained in under a minute on my laptop (2022 MacBook Air with M2 chip, 24Gb RAM). You can use a GPU if you wish, and these are freely available via e.g. Google Colab and Kaggle.

1. Explore the images and metadata. Show examples across people, digits, and sessions. Describe any visible sources of variation and any limitations you see in this simulated data.
2. Define a prediction task in which the response is the writer ID. Build and evaluate a CNN. Report overall writer identification performance and compare performance by digit. Explain what your evaluation measures capture in real-world terms.
3. Build a Siamese network that estimates whether two images were written by the same person. Use contrastive loss or triplet loss. Explain how you construct training examples and evaluate matching performance.
4. Choose and justify your train, validation, and test strategy. Explain what kind of generalization your evaluation represents, and discuss any possible leakage or optimistic performance.
5. Conduct a limited hyperparameter optimization for at least one of the CNN or Siamese models. Assess how sensitive your evaluation measures are to the hyperparameters you choose to vary.
6. Compare the CNN and Siamese results. Critically discuss which digits carry more writer information and how session changes affect performance.

7. Work collaboratively using git and GitHub for the duration of the project, for example with different group members working on the same file, using branches, using pull requests.
8. Write up your work as a report with a maximum word limit of 5000 (excluding AI use statement, appendices, figure captions, and references). The report must contain a plagiarism declaration and an AI use statement (see below).
9. Host your report on a GitHub Pages website, linked to your GitHub repository that you push to programmatically (i.e. not by dragging and dropping files).

Submission deliverables

Submission is via the Assignment tab on the course Amathuba site. **One** member of each group must submit:

1. Submit a reproducible Quarto document containing your project report, describing your approach, results, and conclusions. (one report per group; 50% of final mark).
2. Submit a text file containing (a) the names of all group members, and what their roles were in the group i.e. where you have delegated tasks, what each person did; (b) a link to your GitHub Pages website, (c) a link to a GitHub repository from which your website was built. This should contain the html source files for your website as well as all R source (.R, .qmd, etc) files that contain your written text and code. (one repo and website per group; 10% of final mark).
3. A 20-minute (maximum) group presentation reporting your work. All members of the group must attend the presentation and present some part of the work. (40% of final mark).

Notes

- The dataset contains 25,000 images of size 28×28 : $20 \text{ writers} \times 5 \text{ digits} \times 5 \text{ sessions} \times 50 \text{ images per combination}$.
- Scale pixel values to $[0, 1]$ before fitting a neural network.
- For a same-writer task, positive pairs should show the same writer and negative pairs different writers. Consider whether pair construction should control for digit identity.
- There is no prescribed train/test splits. Your choice and its consequences are part of the assignment.
- Data can be read into R as


```
data <- readRDS("handwriting.rds")
dim(data$images) # 25000 x 28 x 28 x 1
head(data$metadata)
```
- `images` is an integer array with values from 0 (background) to 255 (ink). `metadata` has one row per image and these columns:

Column	Meaning
image_id	Unique row identifier matching the first dimension of <code>images</code>
person_id	Simulated writer, 1–20
digit	Digit represented, 0–4
session	Simulated recording session, 1–5
replicate	Repeated image number within writer, digit, and session, 1–50

- Data can be input to Keras using

```
library(keras3)
use_backend("tensorflow")
x <- data$images / 255
y_person <- data$metadata$person_id - 1L
```

Use of AI

Use of LLMs is **allowed** throughout this assignment.

The weight attached to the group presentation reflects that the work in this assignment is meant to be primarily done by you. Use of LLMs is allowed and encouraged to support your work but you are expected to understand all steps taken and to be able to do these steps independently by the end of the assignment. Sophisticated analyses generated by AI that you do not fully understand and cannot replicate are discouraged and will have a negative impact of your mark.

Use of LLMs in the written and presentation components of this assignment is *allowed but discouraged*. Your written report and your presentation should be *in your own words*. LLMs can be used for proofreading and minor style or presentation issues but should not be used to generate a first draft and text from an LLM should not be used verbatim. There is no real way to enforce this but where excessive AI use is suspected and understanding is not demonstrated in the group presentation, I reserve the right to ask individual groups or students for additional oral or written assessments under closed (to the internet) conditions.

You must submit an AI use statement that describes, for each of the tasks in the section *Tasks* above: (a) what tool you used, (b) what you used it for i.e. specific subtasks, (c) if and how you verified or checked its outputs, and (d) how much of what the AI produced you ended up using (recognizing that dividing up credit between an AI and yourself is an inexact task),

You are responsible for all material you submit, irrespective of whether it originated from you, an LLM or another source.